New Academic Sources Added to the Report

Report: Data Driven Prediction of Local Government Water Service Delivery Outcomes through Machine Learning (N Hlumbene)
File: *Final Report (Chapter 3 revised).docx*. Report page numbers are the printed page numbers in that file as at 1 November 2026.
Each source was checked against the publisher, Crossref, Europe PMC or the paper itself before it was used.

Table 1: Source register

Paper	Year	Author(s)	Themes that make it relevant to this study	Page of the paper referenced, and what it supports	Where it is used in the report (chapter, section, report page)
Random forests	2001	Breiman, L.	Theme 3, machine learning as a predictive tool. Gives the theoretical basis for choosing Random Forest as the main model (empirical research question RQ2).	**Page 5** (abstract): a forest's generalisation error converges to a limit as the number of trees grows, and random feature selection makes it more robust to noise.	Chapter 3, 3.3.3 (p. 35) Chapter 3, 3.4 (p. 37) Chapter 3, 3.6 (p. 39) Chapter 4, 4.2.4 (p. 43) References (p. 66)
Missing value estimation methods for DNA microarrays	2001	Troyanskaya, O., Cantor, M., Sherlock, G., Brown, P., Hastie, T., Tibshirani, R., Botstein, D. & Altman, R.B.	Theme 4, data preparation. The original evaluation of KNN imputation, used to discuss imputation options for incomplete municipal records and to explain why this study chose median imputation with missing value indicators instead (RQ1).	**Page 520** (abstract): KNN imputation was more robust and sensitive than imputation based on singular value decomposition, and both outperformed the row average method.	Chapter 1, Handling Missing Data through Imputation (p. 12) Chapter 3, 3.3.4 (p. 35, cited twice) Chapter 3, 3.4 (p. 37) Chapter 3, 3.6 (p. 39) Chapter 3, 3.7 (p. 40) References (p. 67)
The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets	2015	Saito, T. & Rehmsmeier, M.	Theme 4, model evaluation. Explains why ROC curves alone can mislead when classes are imbalanced, which shapes how the three algorithms are compared (RQ2).	**Page 1** (abstract): ROC plots can be visually deceptive on imbalanced datasets, while precision recall plots give a more accurate indication of future performance.	Chapter 3, 3.3.4 (p. 36, cited twice) Chapter 3, 3.6 (p. 39) Chapter 4, 4.5 (p. 49) References (p. 67)
The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation	2020	Chicco, D. & Jurman, G.	Theme 4, model evaluation. Shows that accuracy and F1 can overstate performance on imbalanced data, which supports reporting several measures (RQ2).	**Page 1** (abstract): accuracy and F1 score can give overoptimistic, inflated results, especially on imbalanced datasets; the authors recommend MCC.	Chapter 3, 3.3.4 (p. 36, cited twice) Chapter 3, 3.3.5 (p. 37) Chapter 3, 3.4 (p. 37) Chapter 3, 3.6 (p. 39) Chapter 3, 3.7 (p. 40)

					Chapter 4, 4.5 (p. 49) References (p. 66)
Leakage and the reproducibility crisis in machine-learning-based science	2023	Kapoor, S. & Narayanan, A.	Theme 4, data preparation. Supports screening out expenditure variables that leak the outcome, as done in Chapter 4, Section 4.3.7 (RQ1 and RQ2).	**Page 1** (summary): data leakage was found in 294 papers across 17 fields using machine learning, sometimes producing overly optimistic conclusions.	Chapter 3, 3.3.4 (p. 35) Chapter 3, 3.3.5 (p. 37) Chapter 3, 3.4 (p. 37) Chapter 3, 3.6 (p. 39) Chapter 3, 3.7 (p. 40) Chapter 4, 4.3.7 (p. 46) Chapter 5, 5.3 (p. 63) References (p. 66)
A unified approach to interpreting model predictions	2017	Lundberg, S.M. & Lee, S.-I.	Theme 4, interpretability, and the conceptual background. The original SHAP paper, used to explain why SHAP is needed to turn Random Forest predictions into risk factors (RQ3).	**Page 4765** (first page and abstract): the most accurate models, such as ensembles, are often hard to interpret, creating a tension between accuracy and interpretability; SHAP assigns each feature an importance value for a particular prediction.	Chapter 3, 3.2.3 (p. 31) Chapter 3, 3.3.4 (p. 36, cited four times) Chapter 3, 3.6 (p. 39) Chapter 4, 4.1 (p. 41) References (p. 66)

Table 2: How each source was verified

Source	Verification
Breiman (2001)	Bibliographic details (Machine Learning, 45(1):5 to 32) confirmed through Crossref for DOI 10.1023/A:1010933404324. Abstract content confirmed through Semantic Scholar.
Troyanskaya et al. (2001)	Details and abstract confirmed on the Oxford Academic (Bioinformatics) article page for DOI 10.1093/bioinformatics/17.6.520.
Saito & Rehmsmeier (2015)	Details and abstract confirmed on the PLoS ONE article page; full text PDF downloaded and checked.
Chicco & Jurman (2020)	Details and abstract confirmed through Europe PMC (PMC6941312) for DOI 10.1186/s12864-019-6413-7.
Kapoor & Narayanan (2023)	Details confirmed through Crossref; summary content confirmed through Europe PMC (PMC10499856), where the article is open access.

Lundberg & Lee (2017)	Details confirmed on the NeurIPS proceedings page; full text PDF downloaded, abstract text checked, and the page range 4765 to 4774 taken from the PDF's metadata.

Reference list entries (Harvard, NWU format, as added to the report)

Breiman, L. 2001. Random forests. *Machine Learning*, 45(1):5–32. https://doi.org/10.1023/A:1010933404324

Chicco, D. & Jurman, G. 2020. The advantages of the Matthews correlation coefficient (MCC) over F1 score and accuracy in binary classification evaluation. *BMC Genomics*, 21:6. https://doi.org/10.1186/s12864-019-6413-7

Kapoor, S. & Narayanan, A. 2023. Leakage and the reproducibility crisis in machine-learning-based science. *Patterns*, 4(9):100804. https://doi.org/10.1016/j.patter.2023.100804

Lundberg, S.M. & Lee, S.-I. 2017. A unified approach to interpreting model predictions. (In *Advances in Neural Information Processing Systems 30 (NIPS 2017)*. p. 4765–4774).

Saito, T. & Rehmsmeier, M. 2015. The precision-recall plot is more informative than the ROC plot when evaluating binary classifiers on imbalanced datasets. *PLoS ONE*, 10(3):e0118432. https://doi.org/10.1371/journal.pone.0118432

Troyanskaya, O., Cantor, M., Sherlock, G., Brown, P., Hastie, T., Tibshirani, R., Botstein, D. & Altman, R.B. 2001. Missing value estimation methods for DNA microarrays. *Bioinformatics*, 17(6):520–525. https://doi.org/10.1093/bioinformatics/17.6.520

Notes

EndNote	Your reference list is an EndNote field. The six entries were typed into it in matching style, and the new in text citations are plain text. Add these sources to your EndNote library before you next use "Update Citations and Bibliography", or EndNote will remove the typed entries.
Dashes	The chapter text contains no dashes. The reference entries keep the page range dash used by your existing references, and the hyphens that are part of the original titles and names (for example "machine-learning-based" and "Su-In"), because Harvard style reproduces titles exactly.
Page numbers	Report page numbers will shift if earlier pages are edited. Recheck them before final submission.